In [94]:
```
import pandas as pd
import numpy as np
from collections import Counter
import matplotlib.pyplot as plt
import seaborn as sns
```

In [95]:
```
# import data
df=pd.read_csv("netflix_titles.csv")
print (f"columns\t: {df.shape[1]}\nrows\t: {df.shape[0]}")
```

```
columns : 12
rows    : 8807
```

In [96]:
```
df.head()
```

Out[96]:

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	NaN	United States	September 25, 2021	2020
1	s2	TV Show	Blood & Water	NaN	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban...	South Africa	September 24, 2021	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi...	NaN	September 24, 2021	2021
3	s4	TV Show	Jailbirds New Orleans	NaN	NaN	NaN	September 24, 2021	2021
4	s5	TV Show	Kota Factory	NaN	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K...	India	September 24, 2021	2021

In [97]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      6173 non-null   object
 4   cast          7982 non-null   object
 5   country       7976 non-null   object
 6   date_added    8797 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8803 non-null   object
 9   duration      8804 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

In [98]:
```
# Clean columns
totalColumn=len(df.columns.tolist())
```

In [99]:
```
# change null data
df['director']=df['director'].fillna("tidak tertera")
df['cast']=df['cast'].fillna('tidak tertera')
df['country']=df['country'].fillna('tidak tertera')
```

In [100…
```
# Assumstion null data is most data
count=Counter(df['date_added'])
new_data=count.most_common(1)[0][0]
df['date_added']=df['date_added'].fillna(new_data)

count=Counter(df['rating'])
new=count.most_common(1)[0][0]
df['rating']=df['rating'].fillna(new)

count =Counter(df['duration'])
new=count.most_common(1)[0][0]
df['duration']=df['duration'].fillna(new)
```

In [101…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      8807 non-null   object
 4   cast          8807 non-null   object
 5   country       8807 non-null   object
 6   date_added    8807 non-null   object
 7   release_year  8807 non-null   int64 
 8   rating        8807 non-null   object
 9   duration      8807 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
dtypes: int64(1), object(11)
memory usage: 825.8+ KB
```

In [102…
```
df['year']=df['date_added'].str.split(',').str[1]
df['datemonth']=df['date_added'].str.split(',').str[0]
df['date']=df['datemonth'].str.split(' ').str[1]
df['month']=df['datemonth'].str.split(' ').str[0]
```

In [103…
```
df["month"].unique().tolist()
count=Counter (df['month'])
most=count.most_common(1)[0][0]
most
df['month']=df['month'].replace(' ',most)
```

In [104…
```
change ={
'September':9,
 'August':8,
 'July':7,
 'June':6,
 'May':5,
 'April':4,
 'March':3,
 'February':2,
 'January':1,
 'December':12,
 'November':11,
 'October':10
}
df['month']=df['month'].replace(change)
```

In [105…
```
len(df['date'])
df['date']=df['date'].replace(' ','12')
```

In [106…
```
# date data
df['month']=df['month'].astype(str)
df['date_added']=df['year']+'-'+df["month"]+'-'+df['date']
df['date_added']=df['date_added'].replace("--","-")
df['date_added'].head()
```

Out[106…
```
0     2021-9-25
1     2021-9-24
2     2021-9-24
3     2021-9-24
4     2021-9-24
Name: date_added, dtype: object
```

In [107…
```
df['date_added'] = (
    df['date_added']
    .astype(str)
    .str.strip()                    # buang spasi depan–belakang
    .str.replace(r'\s+', ' ', regex=True)  # rapihin spasi
    .str.replace('--', '-', regex=False)   # double dash
)
df['date_added'] = pd.to_datetime(
    df['date_added'],
    format='mixed',
    errors='coerce'
)

df['date_added']=pd.to_datetime(df['date_added'])
```

In [108…
```
df['date_added'].head ()
```

Out[108…
```
0   2021-09-25
1   2021-09-24
2   2021-09-24
3   2021-09-24
4   2021-09-24
Name: date_added, dtype: datetime64[ns]
```

In [109…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 16 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   show_id       8807 non-null   object
 1   type          8807 non-null   object
 2   title         8807 non-null   object
 3   director      8807 non-null   object
 4   cast          8807 non-null   object
 5   country       8807 non-null   object
 6   date_added    8807 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64
 8   rating        8807 non-null   object
 9   duration      8807 non-null   object
 10  listed_in     8807 non-null   object
 11  description   8807 non-null   object
 12  year          8807 non-null   object
 13  datemonth     8807 non-null   object
 14  date          8807 non-null   object
 15  month         8807 non-null   object
dtypes: datetime64[ns](1), int64(1), object(14)
memory usage: 1.1+ MB
```

In [110…
```
df['release_year'].head()
```

Out[110…
```
0    2020
1    2021
2    2021
3    2021
4    2021
Name: release_year, dtype: int64
```

In [111…
```
indx=df['release_year']>df['date_added'].dt.year
index_ =df.index[indx].tolist()
for dex in index_:
    future=df.loc[dex,"date_added"]
    future=str(future).split('-')
    newY=df.loc[dex,'release_year'].astype (str)
    month=future[1]
    date=future[2]
    Newdate=newY+'-'+month+'-'+date
    Newdate=pd.to_datetime(Newdate)
    df.loc[dex,'date_added']=Newdate
```

In [112…
```
df.loc [indx,:]
```

Out[112...

	show_id	type	title	director	cast	country	date_added
1551	s1552	TV Show	Hilda	tidak tertera	Bella Ramsey, Ameerah Falzon-Ojo, Oliver Nelso...	United Kingdom, Canada, United States	2021-12-14
1696	s1697	TV Show	Polly Pocket	tidak tertera	Emily Tennant, Shannon Chan-Kent, Kazumi Evans...	Canada, United States, Ireland	2021-11-15
2920	s2921	TV Show	Love Is Blind	tidak tertera	Nick Lachey, Vanessa Lachey	United States	2021-02-13
3168	s3169	TV Show	Fuller House	tidak tertera	Candace Cameron Bure, Jodie Sweetin, Andrea Ba...	United States	2020-12-06
3287	s3288	TV Show	Maradona in Mexico	tidak tertera	Diego Armando Maradona	Argentina, United States, Mexico	2020-11-13
3369	s3370	TV Show	BoJack Horseman	tidak tertera	Will Arnett, Aaron Paul, Amy Sedaris, Alison B...	United States	2020-10-25
3433	s3434	TV Show	The Hook Up Plan	tidak tertera	Marc Ruchmann, Zita Hanrot, Sabrina Ouazani, J...	France	2020-10-11
4844	s4845	TV Show	Unbreakable Kimmy Schmidt	tidak tertera	Ellie Kemper, Jane Krakowski, Tituss Burgess, ...	United States	2019-05-30
4845	s4846	TV Show	Arrested Development	tidak tertera	Jason Bateman, Portia de Rossi, Will Arnett, M...	United States	2019-05-29

	show_id	type	title	director	cast	country	date_added
5394	s5395	Movie	Hans Teeuwen: Real Rancour	Doesjka van Hoogdalem	Hans Teeuwen	Netherlands	2018-07-01
5658	s5659	TV Show	Sense8	tidak tertera	Tuppence Middleton, Brian J. Smith, Doona Bae,...	United States	2018-12-23
5677	s5678	TV Show	Tokyo Trial	tidak tertera	Marcel Hensema, Paul Freeman, Irrfan Khan, Sta...	Canada, Japan, Netherlands	2017-12-13
7063	s7064	Movie	Incoming	Eric Zaragosa	Scott Adkins, Aaron McCusker, Vahldin Prelic, ...	Serbia, United States	2019-10-26
7112	s7113	TV Show	Jack Taylor	Stuart Orme	Iain Glen, Killian Scott, Nora-Jane Noone, Tar...	United States, Ireland	2016-03-31

In [113…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 16 columns):
 #   Column        Non-Null Count  Dtype         
---  ------        --------------  -----         
 0   show_id       8807 non-null   object        
 1   type          8807 non-null   object        
 2   title         8807 non-null   object        
 3   director      8807 non-null   object        
 4   cast          8807 non-null   object        
 5   country       8807 non-null   object        
 6   date_added    8807 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64         
 8   rating        8807 non-null   object        
 9   duration      8807 non-null   object        
 10  listed_in     8807 non-null   object        
 11  description   8807 non-null   object        
 12  year          8807 non-null   object        
 13  datemonth     8807 non-null   object        
 14  date          8807 non-null   object        
 15  month         8807 non-null   object        
dtypes: datetime64[ns](1), int64(1), object(14)
memory usage: 1.1+ MB
```

In [114…
```
df.drop(['year','datemonth','date','month'],axis=1,inplace=True)
```

In [115…
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype         
---  ------        --------------  -----         
 0   show_id       8807 non-null   object        
 1   type          8807 non-null   object        
 2   title         8807 non-null   object        
 3   director      8807 non-null   object        
 4   cast          8807 non-null   object        
 5   country       8807 non-null   object        
 6   date_added    8807 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64         
 8   rating        8807 non-null   object        
 9   duration      8807 non-null   object        
 10  listed_in     8807 non-null   object        
 11  description   8807 non-null   object        
dtypes: datetime64[ns](1), int64(1), object(10)
memory usage: 825.8+ KB
```

In [116…
```
df.head()
```

Out[116…

	show_id	type	title	director	cast	country	date_added	release_year
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	tidak tertera	United States	2021-09-25	2020
1	s2	TV Show	Blood & Water	tidak tertera	Ama Qamata, Khosi Ngema, Gail Mabalane, Thaban…	South Africa	2021-09-24	2021
2	s3	TV Show	Ganglands	Julien Leclercq	Sami Bouajila, Tracy Gotoas, Samuel Jouy, Nabi…	tidak tertera	2021-09-24	2021
3	s4	TV Show	Jailbirds New Orleans	tidak tertera	tidak tertera	tidak tertera	2021-09-24	2021
4	s5	TV Show	Kota Factory	tidak tertera	Mayur More, Jitendra Kumar, Ranjan Raj, Alam K…	India	2021-09-24	2021

In [117…
```
# Data Analyst
df_=df[:]
df_.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8807 entries, 0 to 8806
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype         
---  ------        --------------  -----         
 0   show_id       8807 non-null   object        
 1   type          8807 non-null   object        
 2   title         8807 non-null   object        
 3   director      8807 non-null   object        
 4   cast          8807 non-null   object        
 5   country       8807 non-null   object        
 6   date_added    8807 non-null   datetime64[ns]
 7   release_year  8807 non-null   int64         
 8   rating        8807 non-null   object        
 9   duration      8807 non-null   object        
 10  listed_in     8807 non-null   object        
 11  description   8807 non-null   object        
dtypes: datetime64[ns](1), int64(1), object(10)
memory usage: 825.8+ KB
```

In [118…

```
# realize film
year =df['release_year'].tolist()
count=Counter(year)
print(count)
x=count.keys()
y=count.values()
years=pd.DataFrame({
    'ket':x,
    'values':y
})
```

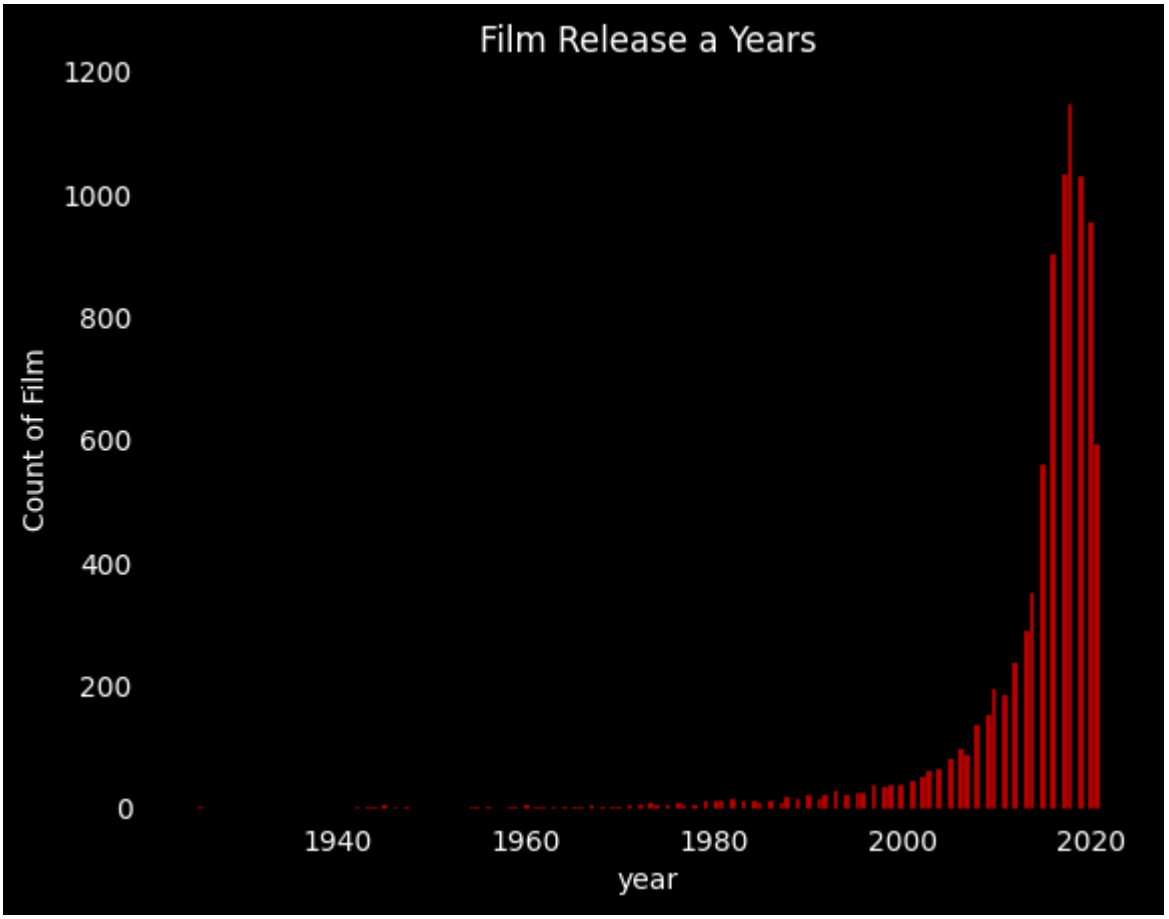
```
Counter({2018: 1147, 2017: 1032, 2019: 1030, 2020: 953, 2016: 902, 2021: 5
92, 2015: 560, 2014: 352, 2013: 288, 2012: 237, 2010: 194, 2011: 185, 2009
: 152, 2008: 136, 2006: 96, 2007: 88, 2005: 80, 2004: 64, 2003: 61, 2002:
51, 2001: 45, 1999: 39, 1997: 38, 2000: 37, 1998: 36, 1993: 28, 1995: 25,
1996: 24, 1992: 23, 1994: 22, 1990: 22, 1988: 18, 1982: 17, 1991: 17, 1989
: 16, 1986: 13, 1981: 13, 1984: 12, 1983: 11, 1980: 11, 1979: 11, 1985: 10
, 1973: 10, 1976: 9, 1987: 8, 1975: 7, 1978: 7, 1974: 7, 1977: 7, 1972: 5,
1971: 5, 1967: 5, 1945: 4, 1960: 4, 1958: 3, 1962: 3, 1968: 3, 1955: 3, 19
44: 3, 1943: 3, 1964: 2, 1954: 2, 1956: 2, 1963: 2, 1970: 2, 1969: 2, 1965
: 2, 1946: 2, 1942: 2, 1961: 1, 1959: 1, 1925: 1, 1966: 1, 1947: 1})
```

In [119…

```
plt.figure(facecolor='black')
graph =sns.histplot(data=df_,x='release_year',color='red')
graph.set_facecolor('black')
plt.title('Film Release a Years',color='white')
sns.despine(right=True, top=True)
plt.xticks(color="white")
plt.yticks(color="white")
plt.xlabel("year",color='white')
plt.ylabel('Count of Film',color='white')
```

Out[119…

```
Text(0, 0.5, 'Count of Film')
```

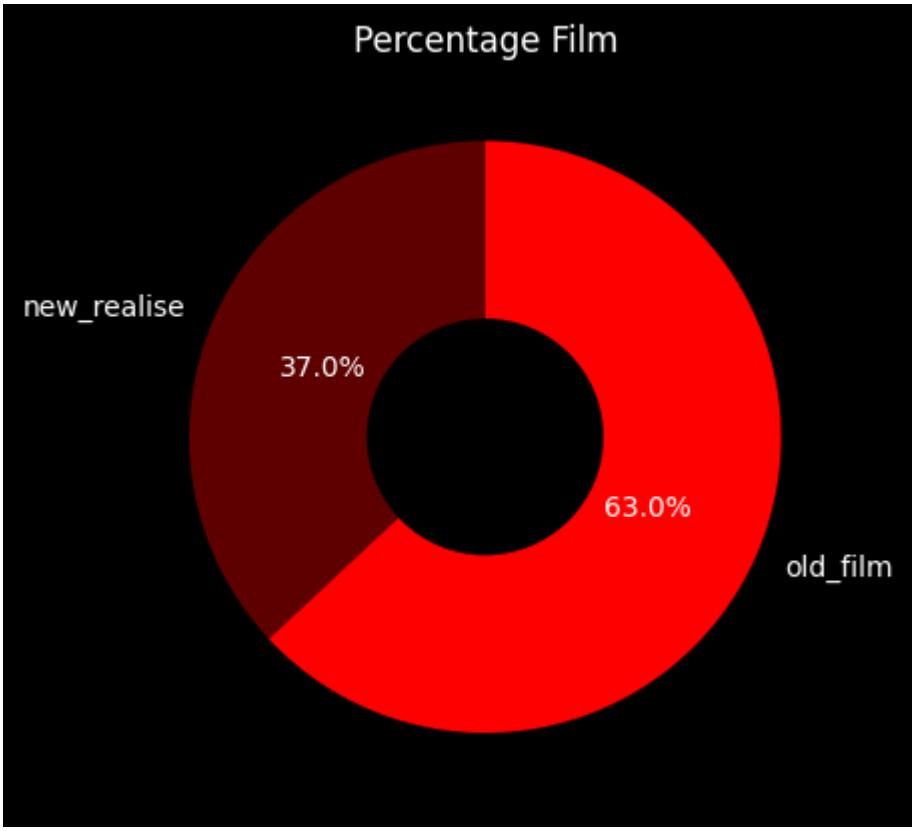


In [120…
```
# Total Film added in Netflix

# minimum film added
year =df_["date_added"].dt.year.tolist()
max_=np.max(year)
min_=np.min(year)
total=len(year)
print (f"From {min_} to {max_}, Netflix added {total} film of the apps")
```

```
From 2008 to 2021, Netflix added 8807 film of the apps
```

In [121…
```
same_realese=(df['release_year']==df['date_added'].dt.year)
tot_same_=len(df.loc[same_realese,'release_year'].tolist())
tot_same_
tot_old_=total-tot_same_
columns=['new_realise','old_film']
values=[tot_same_,tot_old_]
color=['#5f0000','red']
plt.figure(facecolor='black')
plt.pie(values,labels=columns, autopct='%1.1f%%',textprops={"color":'whit
circle=plt.Circle((0,0),0.40,fc='black')
plt.gca().add_artist(circle)
plt.title("Percentage Film",color='white')
plt.show()
```



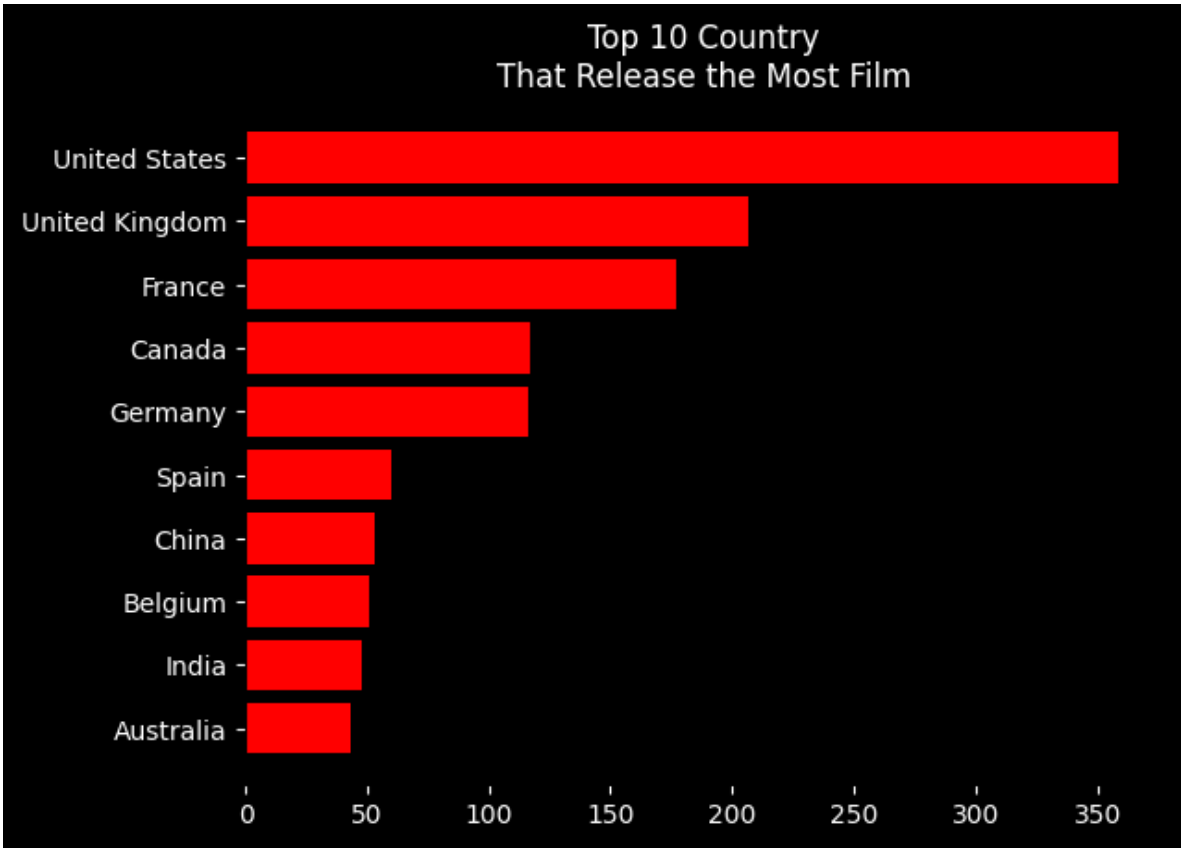
In [126…

```
# data country
count =Counter(df_['country'].unique().tolist())

country=[]
for i in count:
    i=str(i)
    j=i.split(',')
    count =len(j)
    for k in range(count):
       country.append(j[k])

country =[i.strip() for i in country]
count=Counter(country)
most_5=count.most_common(10)

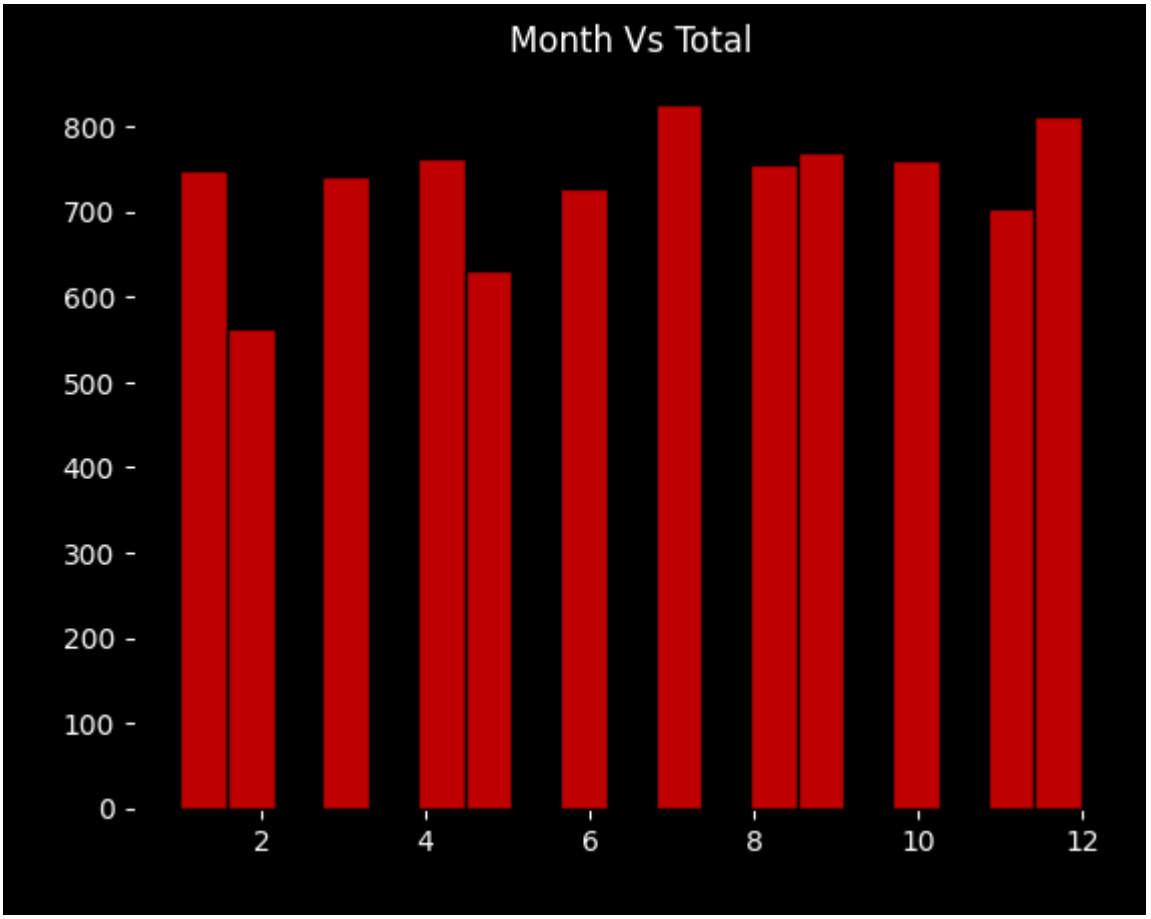
df_tot=pd.DataFrame({
    'country':[most_5[i][0] for i in range(len(most_5))],
    'total':[most_5[i][1] for i in range(len(most_5))]
})
df_tot=df_tot.sort_values(by='total')
plt.figure(facecolor="black")
plt.gca().set_facecolor("black")
graph =plt.barh(df_tot['country'],df_tot['total'],color='red')
plt.tick_params(axis ='x',colors='white')
plt.tick_params(axis ='y',colors='white')
plt.title("Top 10 Country\nThat Release the Most Film", color= 'white')
plt.show()
```



In [123…

```
# Bulan /banyak film
month =df['date_added'].dt.month.tolist()
count =Counter(month)
print (count)
df_month=pd.DataFrame({
    'month':month
})
plt.figure(facecolor='black')
graph=sns.histplot(data=df_month,x ='month',color='red')
graph.set_facecolor('black')
plt.tick_params(axis ='x',colors='white')
plt.tick_params(axis ='y',colors='white')
plt.title("Month Vs Total", color= 'white')
plt.show()
```

```
Counter({7: 827, 12: 813, 9: 770, 4: 764, 10: 760, 8: 755, 1: 748, 3: 742,
6: 728, 11: 705, 5: 632, 2: 563})
```


Month Vs Total
800
700
600
500
400
300
200
100
0
2
4
6
8
10
12